

ON A HYPERBOLIC MODEL FOR HEAT TRANSFER IN COMPRESSIBLE FLUIDS

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Heat conduction in an inviscid compressible flow

Generally described by Euler equation + Fourier's law of heat conduction

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0, \quad (1a)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p(\rho, \eta) \mathbf{I}) = 0, \quad (1b)$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p(\rho, \eta) \mathbf{u} - K \nabla \theta(\rho, \eta)) = 0. \quad (1c)$$

The system describes conservation of mass, momentum and total energy.

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The system describes conservation of mass, momentum and total energy.

Entropy satisfies Clausius-Duhem inequality

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div} \left(\rho \eta \mathbf{u} - \frac{K}{\theta} \nabla \theta \right) = \frac{K}{\theta^2} \|\nabla \theta\|^2 \geq 0.$$

Objective

We would like to provide first-order hyperbolic alternative to the Euler-Fourier system

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0, \quad (2a)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p(\rho, \eta) \mathbf{I}) = 0, \quad (2b)$$

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + p(\rho, \eta) \mathbf{u} - K \nabla \theta(\rho, \eta)) = 0. \quad (2c)$$

Objective properties

We want to obtain a model that satisfies the following properties

- ① Can be derived from a variational principle
- ② First-order hyperbolic system with an explicit eigenstructure
- ③ Total Energy is conserved
- ④ Compatible with the second law of thermodynamics
- ⑤ Gallilean invariant

Plan of presentation

- 1 Model Derivation
- 2 Model analysis and hyperbolicity
- 3 Numerical results

Cattaneo's model

Well-known hyperbolic relaxation of heat equation (1948)

$$\frac{\partial \theta}{\partial t} + \operatorname{div}(\mathbf{q}) = 0$$
$$\tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -K \nabla \theta$$

- No underlying variational principle.
- Not Galilean invariant (fixable)
- Generally considered out of the scope of fluid dynamics.
- when coupled with compressible Euler equations
 - 1 Not hyperbolic in multi-D.
 - 2 Does not satisfy Clausius-Duhem inequality.

Cattaneo's model - No entropy

$$\rho_t + (\rho u)_x = 0$$

$$(\rho u)_t + (\rho u^2 + p)_x = 0$$

$$\left(\rho \left(\varepsilon + \frac{1}{2} u^2 \right) \right)_t + \left(\rho u \left(\varepsilon + \frac{1}{2} u^2 \right) \right)_x = -(p u)_x - q_x$$

$$\tau q_t + \tau u q_x + q = -\kappa \theta_x$$

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$$\tau q_t + \tau u q_x + q = -\kappa \theta_x$$

The Gibbs identity implies that

$$\rho \theta \frac{d\eta}{dt} = -q_x$$

which can be cast in conservative form as

$$(\rho \eta)_t + \left(\rho \eta u + \frac{q}{\theta} \right)_x = -\frac{q \theta_x}{\theta^2} \geq? 0$$

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$$(\rho \eta)_t + \left(\rho \eta u + \frac{q}{\theta} \right)_x = -\frac{q \theta_x}{\theta^2} \stackrel{?}{>} 0$$

not necessarily satisfied because $q \neq -K \theta_x \quad \forall \tau > 0$.

Entropy inequality: Often missing

1. Cimmelli VA, Kosinski W, Saxton K. 1991 A new approach to the theory of heat conduction with finite wave speeds. *Le Matematiche* 46, 95–105.
2. Rubin M. 1992 Hyperbolic heat conduction and the second law. *Int. J. Eng. Sci.* 30, 1665–1676.
3. Christov C. 2009 On frame indifferent formulation of the Maxwell–Cattaneo model of finite-speed heat conduction. *Mech. Res. Commun.* 36, 481–486.
4. Angeles F. 2022 Non-hyperbolicity of the inviscid Cattaneo–Christov system for compressible fluid flow in several space dimensions. *Quart. J. Mech. Appl. Math.* 75, 147–170.
5. etc .

About Euler-Lagrange equations

Given a Lagrangian, you can derive the Euler-Lagrange equation

$$\mathcal{L}(q, \dot{q}, \nabla q) \implies \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla q} \right) = \frac{\partial \mathcal{L}}{\partial q}$$

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Things are already more complicated for Euler equations

$$\mathcal{L}(\rho, \mathbf{u}) = \int_{\Omega_t} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon(\rho, \eta) \right) d\Omega,$$

$$\delta \rho = -\operatorname{div}(\rho \delta \mathbf{x}), \quad \delta \mathbf{u} = \frac{\partial \delta \mathbf{x}}{\partial t} + \frac{\partial \delta \mathbf{x}}{\partial \mathbf{x}} \mathbf{u} - \frac{\partial \mathbf{u}}{\partial \mathbf{x}} \delta \mathbf{x}$$

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After a bit of calculus $\implies \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon}{\partial \rho} \mathbf{I} \right) = 0$

Thermal displacement (Green-Naghdi 1991)

In this paper :

[1] Green, A. E., & Naghdi, P. (1991). A re-examination of the basic postulates of thermomechanics. Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences, 432(1885), 171-194.

The authors introduce an independent auxiliary potential $\phi(\mathbf{x}, t)$ as a thermal analogue of the kinematic variables such that

$$\dot{\phi}(\mathbf{x}, t) = -\theta(\mathbf{x}, t)$$

One can then write the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^*(\rho, \dot{\phi}) \right) d\Omega,$$

where

$$\varepsilon(\rho, \eta) = \varepsilon^*(\rho, \dot{\phi}) - \eta \dot{\phi}, \quad \text{with} \quad \eta = \frac{\partial \varepsilon^*}{\partial \dot{\phi}}.$$

Entropy as an Euler-Lagrange equation

Given the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^{\star}(\rho, \dot{\phi}) \right) d\Omega, \quad \left(\dot{\phi} = \frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi \right)$$

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One obtains

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} \mathbf{I} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \mathbf{x})$$

$$\frac{\partial}{\partial t} \left(\rho \frac{\partial \varepsilon^*}{\partial \dot{\phi}} \right) + \operatorname{div} \left(\rho \frac{\partial \varepsilon^*}{\partial \dot{\phi}} \mathbf{u} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \phi)$$

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One obtains

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div} \left(\rho \mathbf{u} \otimes \mathbf{u} + \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} \mathbf{I} \right) = 0, \quad (\text{Euler-Lagrange for } \delta \mathbf{x})$$

$$\frac{\partial}{\partial t} (\rho \eta) + \operatorname{div} (\rho \eta \mathbf{u}) = 0, \quad (\text{Euler-Lagrange for } \delta \phi)$$

$$\frac{\partial \rho}{\partial t} + \operatorname{div} (\rho \mathbf{u}) = 0 \quad (\text{Constraint})$$

- A similar idea was also used in Lagrangian coordinates in *Peshkov, Pavelka, Grmela and Romenski (2018)*.

Extension of Green-Naghdi's philosophy

Consider the Lagrangian

$$\mathcal{L}(\rho, \mathbf{u}, \nabla \phi, \dot{\phi}) = \int_{\Omega} \left(\frac{1}{2} \rho \|\mathbf{u}\|^2 - \rho \varepsilon^{\star}(\rho, \dot{\phi}) - \frac{1}{2} \alpha(\rho) \|\nabla \phi\|^2 \right) d\Omega,$$

where the function $\alpha(\rho)$ is an arbitrary positive function of density.

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$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I} + \alpha(\rho) \nabla \phi \otimes \nabla \phi) = 0,$$

$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \nabla \phi) = 0,$$

where $P = \rho^2 \frac{\partial \varepsilon^*}{\partial \rho} + \frac{1}{2}(\rho \alpha'(\rho) - \alpha(\rho)) \|\nabla \phi\|^2$

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✗ The PDE is of second order and depends on $\nabla \phi$.

Solution: First-order reduction

Let us introduce a new variable $\mathbf{j}$ defined as

$$\mathbf{j} = \nabla \phi$$

We then take the material derivative of this equation

$$\mathcal{D}_t \mathbf{j} = \nabla \phi_t + \frac{\partial \nabla \phi}{\partial \mathbf{x}} \cdot \mathbf{u} \quad (3)$$

which we rewrite as

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla (\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + (\nabla \wedge \mathbf{j}) \wedge \mathbf{u} = 0 \quad (4)$$

Dissipationless system of equations

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + \Pi) = 0, \quad \Pi = P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}$$

$$\frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} = 0,$$

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$$\frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) = 0.$$

Total energy conservation is obtained as a consequence

$$\frac{\partial E}{\partial t} + \operatorname{div}(E \mathbf{u} + \Pi \mathbf{u} + \mathbf{q}) = 0, \quad \mathbf{q} = \alpha(\rho) \theta(\rho, \eta) \mathbf{j}$$

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Additional term in the energy conservation *should* be the heat flux.

Rayleigh dissipation function

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) &= 0, \\ \frac{\partial \rho \mathbf{u}}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P(\rho, \eta, \mathbf{j}) \mathbf{I} + \alpha(\rho) \mathbf{j} \otimes \mathbf{j}) &= 0, \\ \frac{\partial \mathbf{j}}{\partial t} + \nabla(\mathbf{j} \cdot \mathbf{u} + \theta(\rho, \eta)) + \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} - \left(\frac{\partial \mathbf{j}}{\partial \mathbf{x}} \right)^T \right) \mathbf{u} &= -\frac{\partial \mathcal{R}}{\partial \mathbf{j}}, \\ \frac{\partial \rho \eta}{\partial t} + \operatorname{div}(\rho \eta \mathbf{u} + \alpha(\rho) \mathbf{j}) &= \frac{\alpha(\rho)}{\theta(\rho, \eta)} \frac{\partial \mathcal{R}}{\partial \mathbf{j}} \cdot \mathbf{j}.\end{aligned}$$

Here $\mathcal{R}$ is the *Rayleigh dissipation* function and which we take in the simplest form as

$$\mathcal{R} = \frac{1}{2\tau} \|\mathbf{j}\|^2, \quad \frac{\partial \mathcal{R}}{\partial \mathbf{j}} = \frac{1}{\tau} \mathbf{j}$$

Asymptotic Analysis and compatibility with Fourier's law

we can expand the variables in power series of τ

$$\rho = \rho_0 + O(\tau), \quad \mathbf{u} = \mathbf{u}_0 + O(\tau), \quad \eta = \eta_0 + O(\tau), \quad \mathbf{j} = \mathbf{j}_0 + \tau \mathbf{j}_1 + o(\tau),$$

and we focus on the $\mathbf{j}$ equation

$$\tau \left(\frac{\partial \mathbf{j}_0}{\partial t} + \frac{\partial \mathbf{j}_0}{\partial \mathbf{x}} \mathbf{u}_0 + \left(\frac{\partial \mathbf{u}_0}{\partial \mathbf{x}} \right)^T \mathbf{j}_0 + \nabla \theta(\rho_0, \eta_0) \right) = -(\mathbf{j}_0 + \tau \mathbf{j}_1) + o(\tau).$$

to obtain

$$\mathbf{j}_0 = 0, \quad \mathbf{j}_1 = -\nabla \theta(\rho_0, \eta_0), \quad \implies \boxed{\mathbf{j} = -\tau \nabla \theta(\rho_0, \eta_0) + o(\tau).}$$

Under these considerations, the heat flux vector expresses as

$$\mathbf{q} = -\tau \alpha(\rho_0) \theta(\rho_0, \eta_0) \nabla \theta(\rho_0, \eta_0).$$

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and we focus on the $\mathbf{j}$ equation

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Under these considerations, the heat flux vector expresses as

$$\mathbf{q} = -\tau \alpha(\rho_0) \theta(\rho_0, \eta_0) \nabla \theta(\rho_0, \eta_0).$$

compatible with Fourier's law if

$$\tau = \frac{K}{\alpha(\rho_0) \theta(\rho_0, \eta_0)}$$

Energy convexity

Total energy is given by

$$E(\rho, \mathbf{m}, s, \mathbf{j}) = \frac{1}{2\rho} \|\mathbf{m}\|^2 + \rho \varepsilon(\rho, s/\rho) + \frac{1}{2} \alpha(\rho) \|\mathbf{j}\|^2, \quad \mathbf{m} = \rho \mathbf{u}, s = \rho \eta$$

Sufficient criterion for energy convexity

$$\text{if } \frac{\partial^2}{\partial \rho^2} \left(\frac{1}{\alpha(\rho)} \right) \leq 0, \quad \text{for } \rho > 0.$$

then E is also a convex function of $\mathbf{Q}$.

We choose a simple function fitting this criterion

$$\alpha(\rho) = \frac{\varkappa^2}{\rho}, \quad \varkappa = \text{cst.}$$

Energy convexity

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$$\alpha(\rho) = \frac{\varkappa^2}{\rho}, \quad \varkappa = cst.$$

(Another possibility is $\alpha(\rho) = cst$, taken in *Peshkov et.al. (2018)*)

Hyperbolicity

system can be cast into quasilinear form

$$\frac{\partial \mathbf{V}}{\partial t} + \mathbf{A}(\mathbf{V}) \frac{\partial \mathbf{V}}{\partial \mathbf{x}} = 0$$

where $\mathbf{A}$ admits 8 eigenvalues whose expressions are given by

$$\left\{ \begin{array}{l} \chi_1 = u_1 - \sqrt{Z_1 + Z_2}, \\ \chi_2 = u_1 - \sqrt{Z_1 - Z_2}, \\ \chi_{3-6} = u_1, \\ \chi_7 = u_1 + \sqrt{Z_1 - Z_2}, \\ \chi_8 = u_1 + \sqrt{Z_1 + Z_2} \end{array} \right. \quad \text{where} \quad \left\{ \begin{array}{l} Z_1 = \frac{1}{2} (a_p^2 + a_T^2 + a_j^2), \\ Z_2 = \sqrt{a_{pT}^4 + \frac{1}{4} (a_p^2 - a_T^2)^2}, \\ a_p^2 = \frac{\partial p}{\partial \rho}, \quad a_T^2 = \frac{\kappa^2}{\rho^2} \frac{\partial \theta}{\partial \eta}, \\ a_{pT}^4 = \frac{\kappa^2}{\rho^2} \frac{\partial p}{\partial \eta} \frac{\partial \theta}{\partial \rho}, \quad a_j^2 = \frac{2\kappa^2}{\rho^2} (j_2^2 + j_3^2). \end{array} \right.$$

1D case: Rankine-Hugoniot conditions

In one dimension of space the RH conditions write

$$\begin{aligned} [\mathcal{M}] &= 0, \\ \left[p + \frac{\mathcal{M}^2}{\rho} \right] &= 0, \\ \left[\mathcal{M} \left(\frac{\mathcal{M}^2}{2\rho^2} + \varepsilon + \frac{p}{\rho} + \frac{1}{2} \frac{\kappa^2}{\rho^2} j^2 \right) + \frac{\kappa^2}{\rho} \theta j \right] &= 0, \\ \left[\mathcal{M} \frac{j}{\rho} + \theta \right] &= 0, \end{aligned}$$

where we have defined the mass flux across the discontinuity front by $\mathcal{M} = \rho(u - \mathcal{D})$ and $\mathcal{D}$ is the discontinuity speed.

Non-existence of contact discontinuity

On contact discontinuities, that is for $\mathcal{M} = 0$, one obtains by direct substitution

$$[p] = 0, \quad \left[\kappa^2 \frac{j}{\rho} \theta \right] = 0, \quad [\theta] = 0.$$

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Since p and θ are continuous across the discontinuity, the density will be as well.

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Since p and θ are continuous across the discontinuity, the density will be as well. Thus:

$$[\rho] = 0, \quad [u] = 0, \quad [\eta] = 0, \quad [j] = 0,$$

Therefore solution is continuous: no contact discontinuities are admissible in this case.

Hugoniot curve for hyperbolic heat conductivity

The equations further simplify into

$$\left[\left(\frac{\mathcal{M}^2}{2\rho^2} + \frac{p}{\rho} + \varepsilon \right) - \frac{\varkappa^2}{2} \frac{\theta^2}{\mathcal{M}^2} \right] = 0, \quad (6)$$

with

$$\mathcal{M}^2 = -\frac{[p]}{[v]}, \quad v = \frac{1}{\rho}. \quad (7)$$

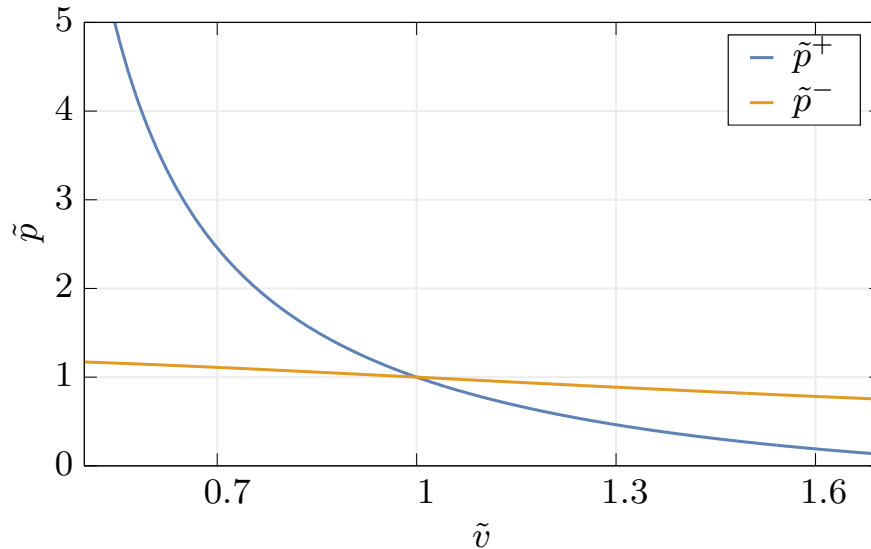
This finally allows us to obtain from equation (6) the following consequence of the Rankine–Hugoniot conditions

$$\varepsilon_R - \varepsilon_L + \frac{1}{2}(p_R + p_L)(v_R - v_L) + \frac{\varkappa^2}{2}(v_R - v_L) \frac{\theta_R^2 - \theta_L^2}{p_R - p_L} = 0. \quad (8)$$

Definition We call *Hugoniot curve* with center (v_0, p_0) the curve in the (v, p) - plane defined as

$$\varepsilon - \varepsilon_0 + \frac{1}{2}(p + p_0)(v - v_0) + \frac{\varkappa^2}{2} \frac{(\theta^2 - \theta_0^2)(v - v_0)}{(p - p_0)} = 0. \quad (9)$$

Hugoniot Locus (polytropic gas equation of state)



Study of the Hugoniot curves shows interesting possible solutions:

- Expansion shocks,
- Compression fans,
- Compound shocks.

Clausius-Duhem across shocks

Through the shock, the jump relations are supplemented by the Clausius-Duhem inequality

$$\left[\mathcal{M}\eta + \frac{\varkappa^2}{\rho} j \right] \geq 0. \quad (10)$$

If the subscript 0 denotes the state ahead of the shock front, then this inequality can be rewritten as

$$\Psi(v) = \eta - \eta_0 - \frac{\varkappa^2}{\mathcal{M}^2}(\theta - \theta_0) \geq 0, \quad v = \frac{1}{\rho}. \quad (11)$$

This inequality can be seen as a criterion of admissibility of shock waves for our system. It should be checked for each branch of the Hugoniot curve (9).

Recovery of Fourier law: Shock tube problem

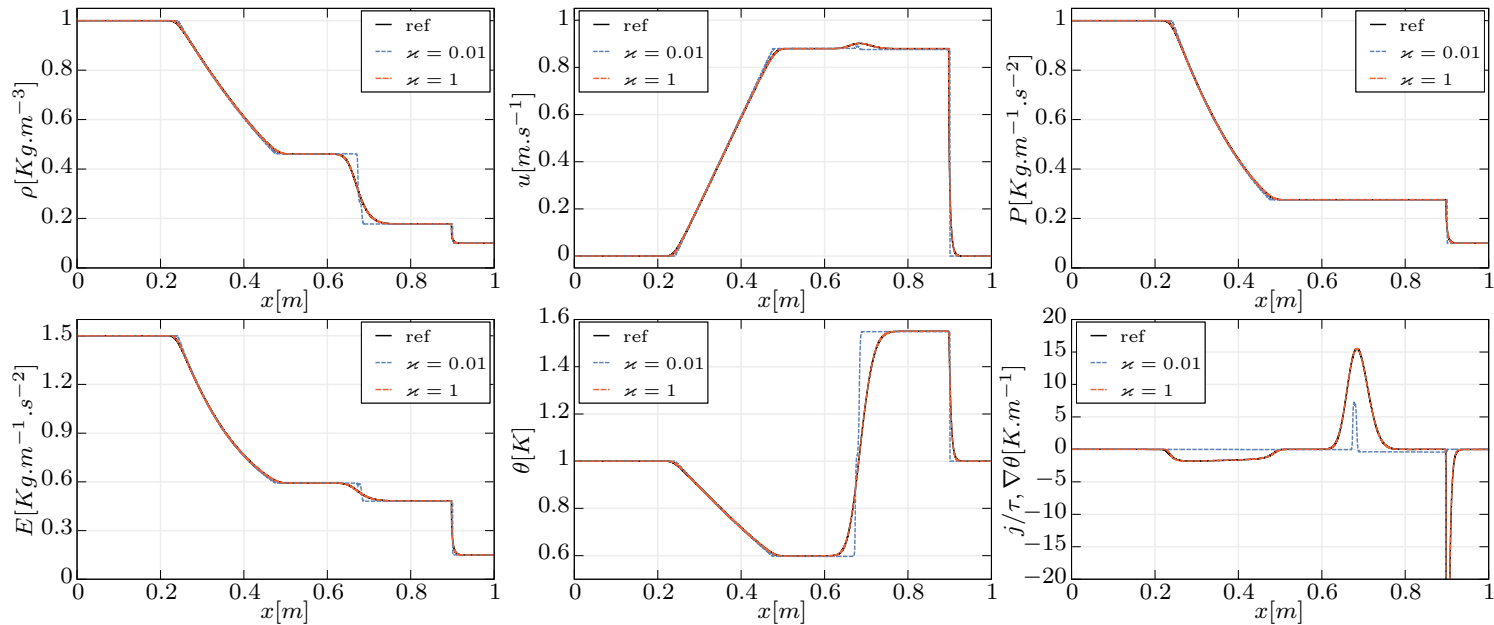


Figure 1: Shock tube with heat conduction. The solution is given at final time $t = 0.2$. Parameters: $\text{CFL} = 0.9$, $\gamma = 5/3$, $c_V = 3/2$, $K = 10^{-3}$. Relaxation time is taken as $\tau = \frac{K}{\alpha(\rho_0)\theta(\rho_0, \eta_0)}$

Recovery of Fourier law: Shock tube problem - Zoom

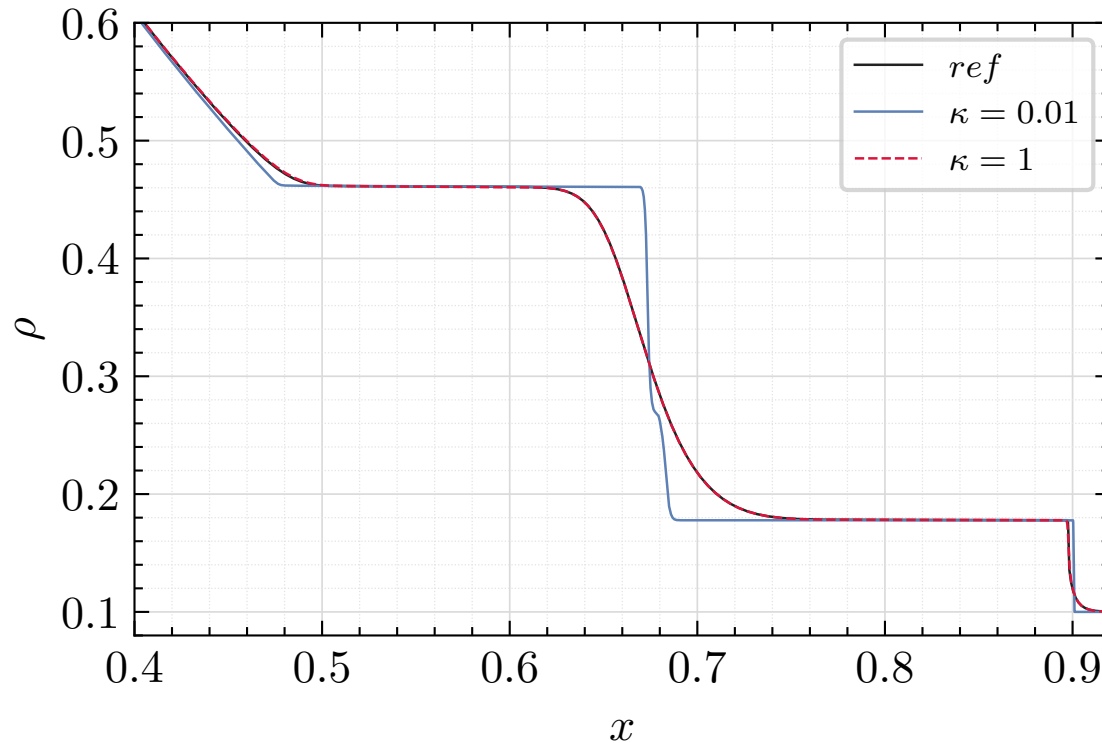


Figure 2: Zoom on the diffuse 'contact discontinuity'

Expansion shock solution

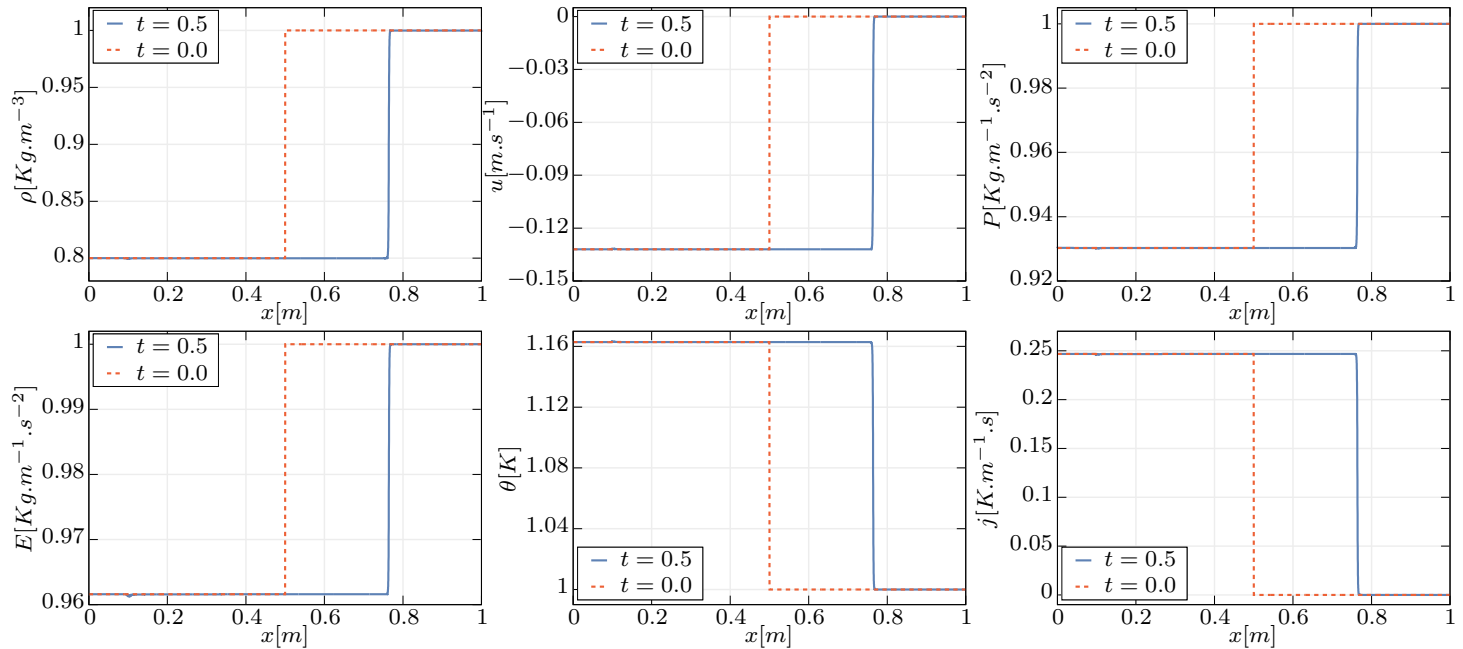


Figure 3: Numerical result for an expansion shock solution on the computational domain $[0, 1]$, discretized over $N = 10000$ cells displayed at final time $t = 0.5$. Parameters: $\text{CFL} = 0.9$, $\gamma = 2$, $c_V = 1$, $\varkappa = 0.8$.

Compression fan

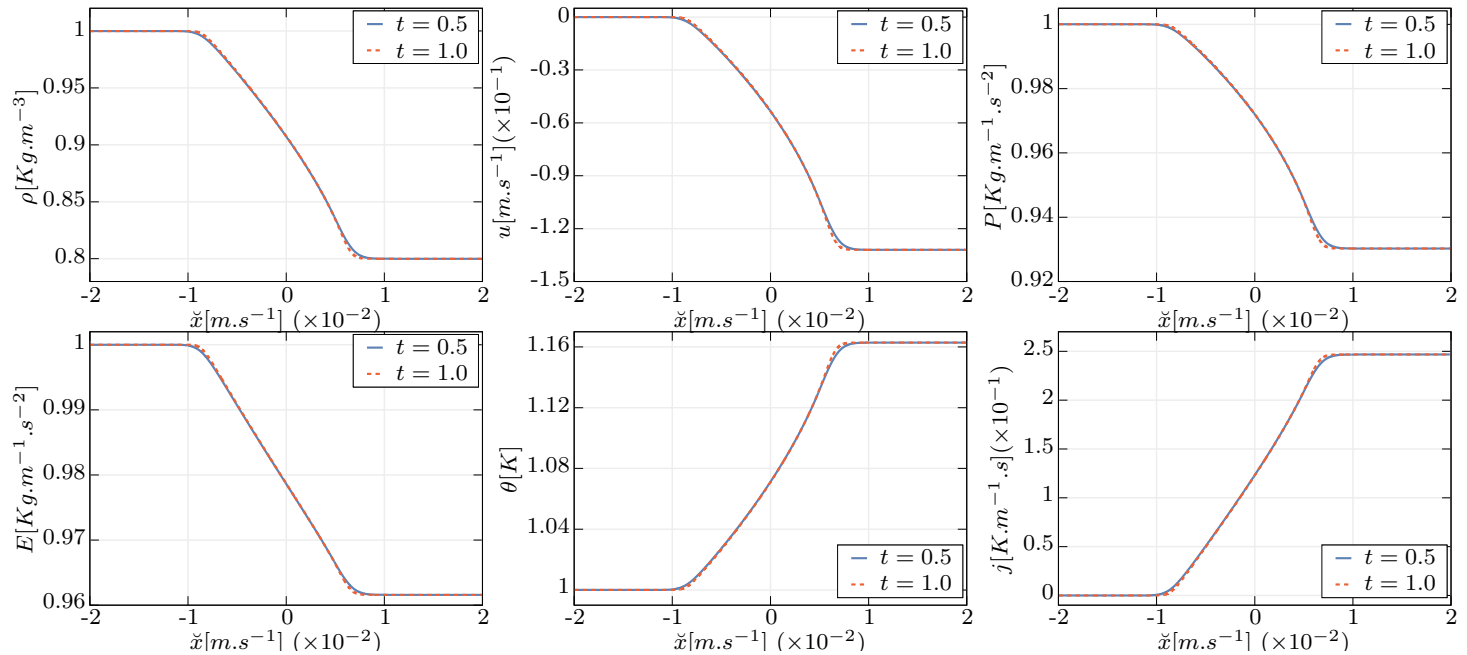


Figure 4: Numerical result for a compression fan solution. Parameters: $CFL = 0.9$, $\gamma = 2$, $c_V = 1$, $\varkappa = 0.8$

Compound shock solution

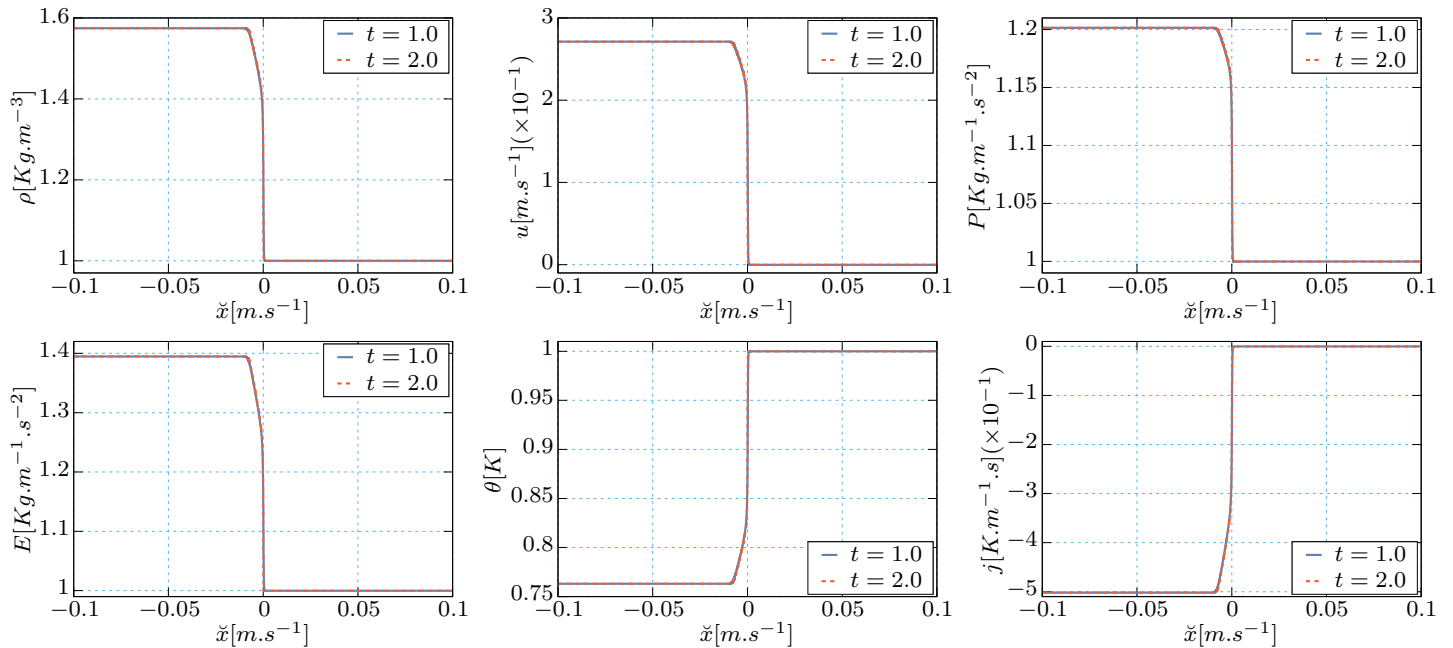


Figure 5: Compound shock plotted as a function of the self-similar coordinate $\check{x} = (x - \mathcal{D}_\star t)/t$. CFL = 0.9, $\gamma = 2$, $c_V = 1$, $\varkappa = 1.3$.

Multi-D result (By W.Boscheri 2025)

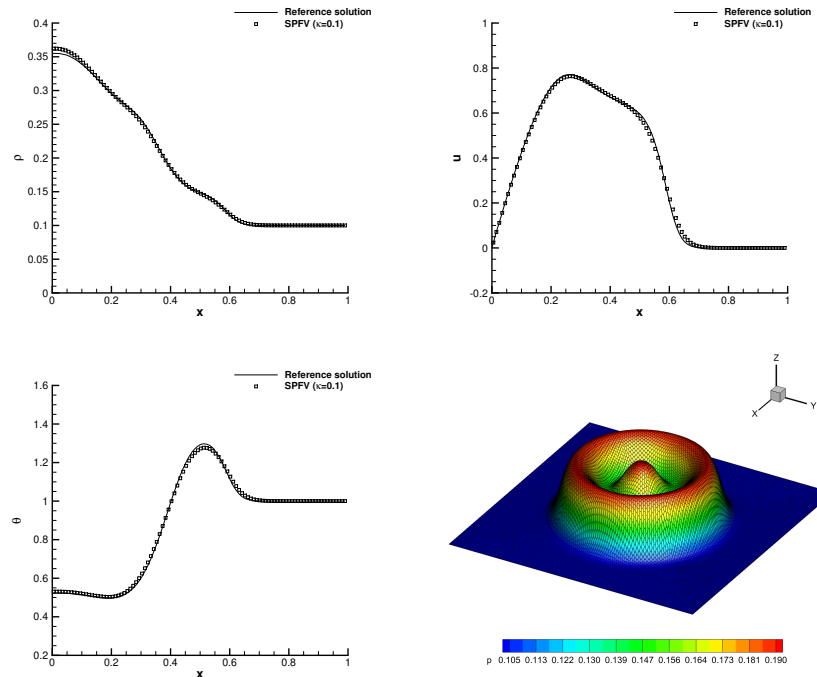


Figure 6: Explosion problem at time $t = 0.2$ with 200 equidistant points is used for the SPFV solution. 3D view, Voronoi mesh and color map of the pressure distribution are shown in the bottom right panel.

Conclusion and Perspectives

- Heat conduction can be modeled by hyperbolic equations derived from variational principles.
- Entropy equation can be derived as an Euler-Lagrange equation.
- Fourier's law can be obtained as asymptotic behavior.
- The study of the Rankine Hugoniot conditions reveals interesting solutions.

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Perspectives

- Rigorous Justification of the relaxation limit

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Perspectives

- Rigorous Justification of the relaxation limit
- Comparison with kinetic theory models.
- Complete study of the Riemann problem.
- Extension to more general heat conduction.

Thank you for your attention !

[1] Dhaouadi, Firas, and Sergey Gavriluk. "An Eulerian hyperbolic model for heat transfer derived via Hamilton's principle: analytical and numerical study." Proceedings of the Royal Society A 480.2283 (2024): 20230440.

And references therein.

Code is published alongside the paper.

Kinetic theory of rarefied gases : Bhatnagar-Gross-Krook (BGK) model

Kinetic model (cf. W. Haegeman, G. Orlando, M. Massot, S. Kokh (2024)) :

$$f_t + \mathbf{v} \cdot \nabla_{\mathbf{x}} f = \frac{M[f] - f}{\tau}.$$

Eight-moments system

$$\rho_t + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$(\rho \mathbf{u})_t + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = 0,$$

$$(\rho E)_t + \operatorname{div}((\rho E + p) \mathbf{u} + \mathbf{q}) = 0,$$

$$\begin{aligned} \mathbf{q}_t + (\mathbf{u} \cdot \nabla) \mathbf{q} + \frac{2\gamma - 1}{\gamma} \left(\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \operatorname{div}(\mathbf{u}) \mathbf{I} \right) \mathbf{q} + \frac{\gamma - 1}{\gamma} \left(\frac{\partial \mathbf{u}}{\partial \mathbf{x}} \right)^T \mathbf{q} \\ + \frac{\gamma p}{\gamma - 1} \nabla \left(\frac{p}{\rho} \right) = -\frac{\mathbf{q}}{\tau}, \quad \gamma = \frac{5}{3}. \end{aligned}$$

Compound shocks

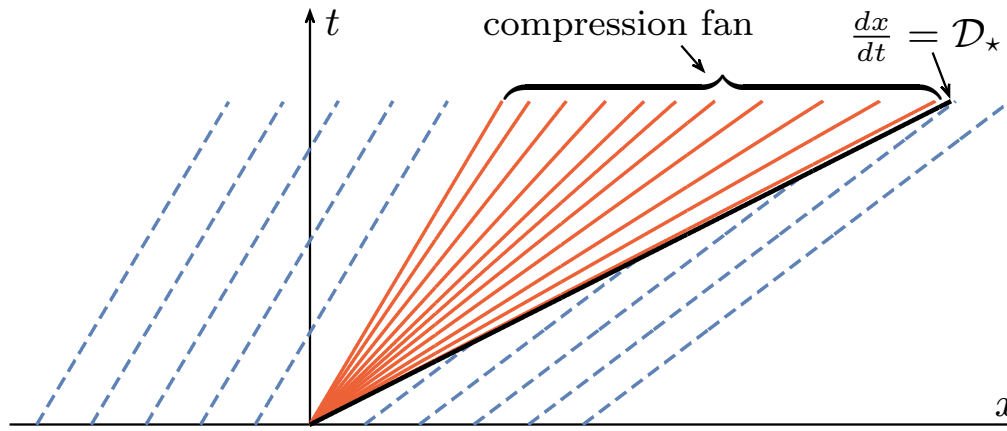


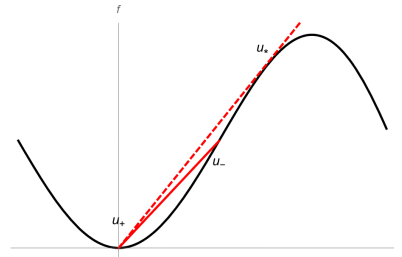
Figure 7: Schematic representation of the wave pattern in the $x - t$ plane, for a compound shock solution. The shock propagates to the right, followed by a right facing compression fan.

Oleinik-Liu admissibility criterion

O. Oleinik (scalar conservation law), B. Wendroff (Euler equations with $p_v < 0$), T.- P. Liu (systems of conservation laws).=====

Scalar conservation laws :

$$u_t + f(u)_x = 0.$$



The shock is *admissible* iff

$$D = \frac{f(u_-) - f(u_+)}{u_- - u_+} \geq \frac{f(u) - f(u_+)}{u - u_+}$$

for any u in the interval (u_-, u_+)